



JOHNS HOPKINS
CAREY BUSINESS SCHOOL

Lecture 6

BU.330.760 Generative AI for Business

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Text Tasks Wrap-up

- » “LLM” now include both language and reasoning models
- » Ways you can use LLM
 - Train your own model, if enough data and enough infrastructure ✓
 - Fine-tune an existing model, need data and infrastructure as well ✓
 - RAG, relatively cheaper but need security control ✓
 - Prompt engineering, cheapest and can be powerful given larger context window ✓
- » *Then why do companies still want to train their own models?*



Issues in Current Mechanism

- » Attention can be distracted
- » Reinforcement learning can be manipulated and misleading ζ
- » Chain-of-thought can be hacked—
- » Or more fundamentally, **language->information->physical world**



Agents are the Future

» Agentic AI is the current trend

» *“What is your process?”*

» Guest speaker sessions of online section open to on-site students

» Recordings posted on Canvas



Announcements

- » Sample assignments posted on Canvas
- » Office hours this week will be by appointment
 - No Wednesday regular office hours
- » Recent job openings at TikTok ([Links deleted 4/29/2025](#))



Today's Agenda

» Image Models

- Generative Adversarial Networks (GAN)
- Diffusion

» Multimodality Models

- DALL.E
- Stable Diffusion

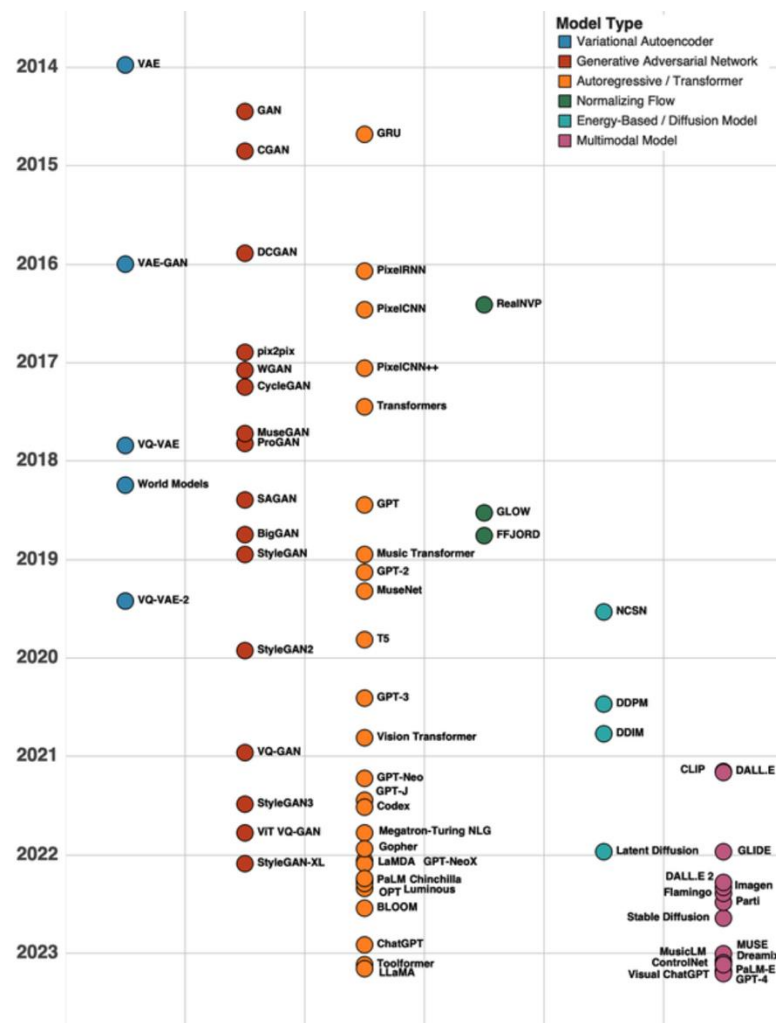


» Dark Sides

» Music Models (very briefly)

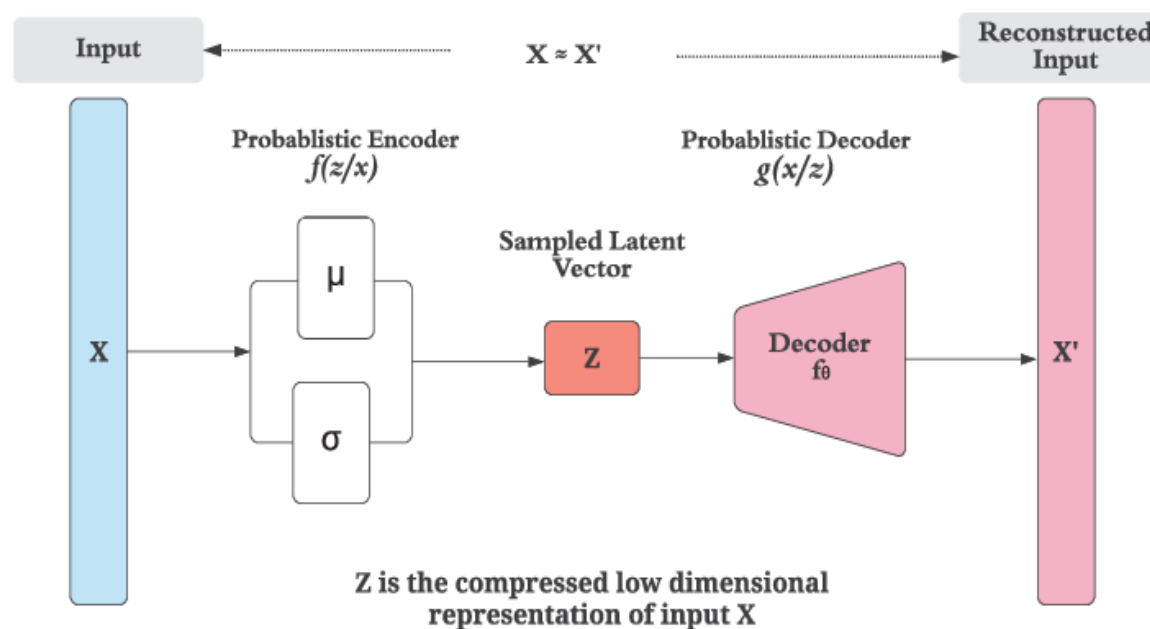
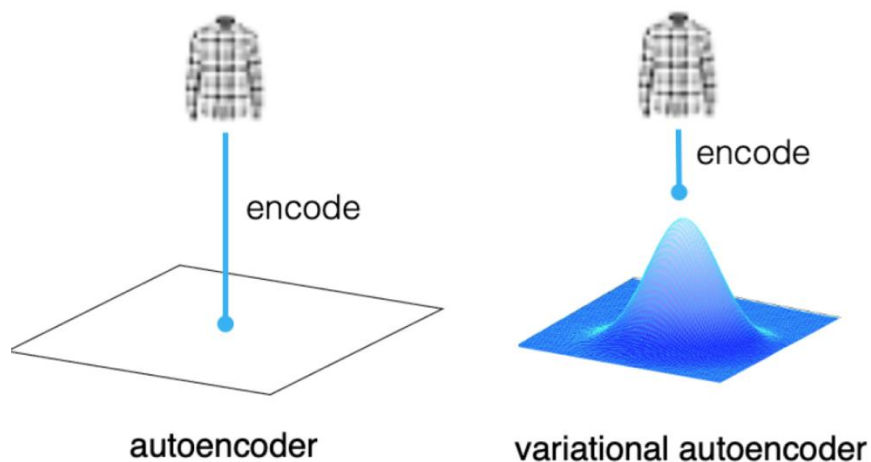
- Transformer
- MuseGAN

Model History (Recap)



VAE (Recap)

- » Variational autoencoder: each data point is mapped to a multivariate normal distribution around a point in the latent space
- » Language is discrete, but images are continuous



GAN History



» GAN: Generative Adversarial Networks

» **GAN**: 2014, Goodfellow et al. presented the model at the Neural Information Processing Systems conference (NeurIPS) in Montreal

- A key turning point in the history of generative modeling

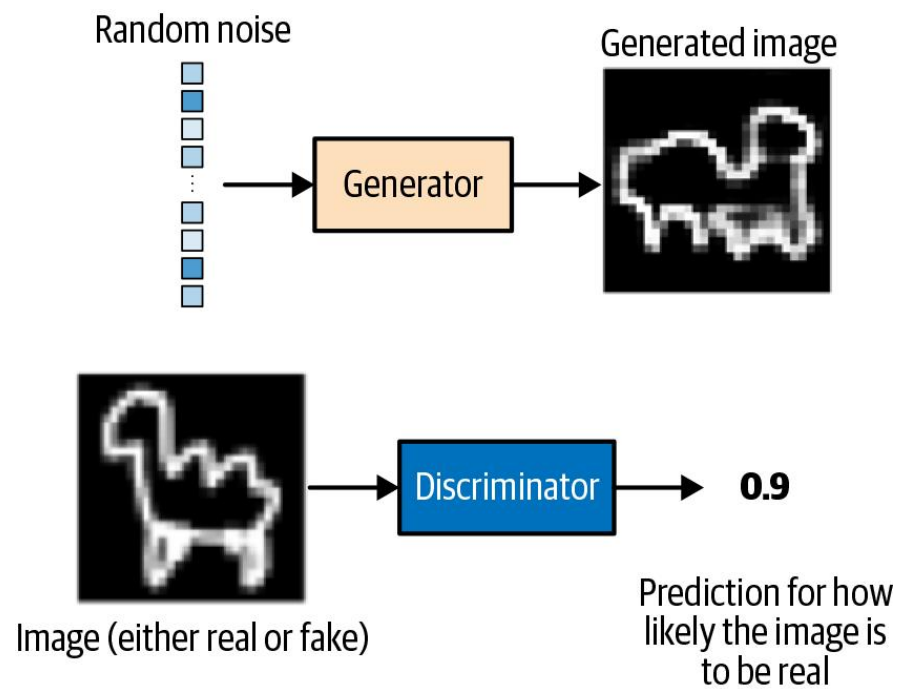
» **Deep Convolutional GAN**: 2015, “Unsupervised Representation Learning with Deep Convolutional Generative Adversarial Networks”

Design Concept



- » Two neural network models contest with each other
 - A battle between two adversaries!
- » A **generative model** that captures the distribution of the training set and generate samples
 - Convert random noise into observations that look as if they have been sampled from the original dataset
- » A **discriminative model** that estimates the probability (predict) that a sample came from the training data and not the generative model above
- » *Is the idea similar to something we've learned?*

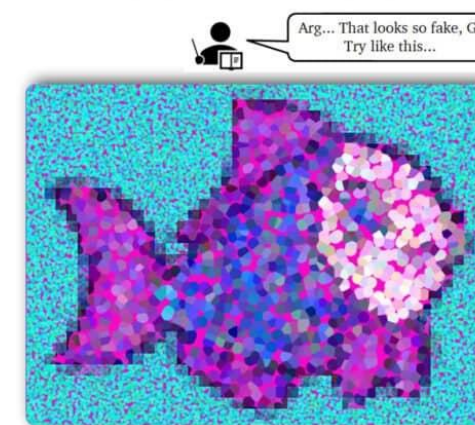
Illustration



Generator



Discriminator



Source: <https://learnopencv.com/introduction-to-generative-adversarial-networks/>



Training: High Level

- » Generator and discriminator engage in a back-and-forth competition
 - Generator strives to create data that is increasingly difficult for the discriminator to distinguish
 - Discriminator aims to improve its ability to distinguish
 - *Why need to improve discriminator?*

- » Once training is complete, the generator can produce data that is remarkably realistic and challenging to differentiate from real data

Training Process

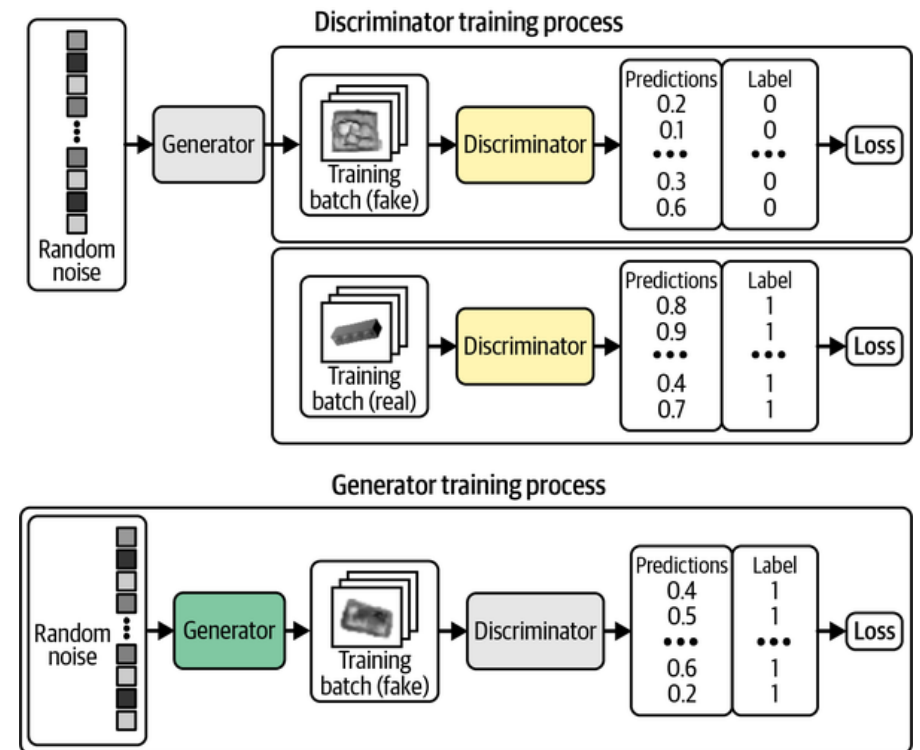
» Discriminator:

- Training set of some fake images and some real images
- Supervised learning with labels

» Generator:

- Generate images
- Get scores from discriminator
- Compare to label 1

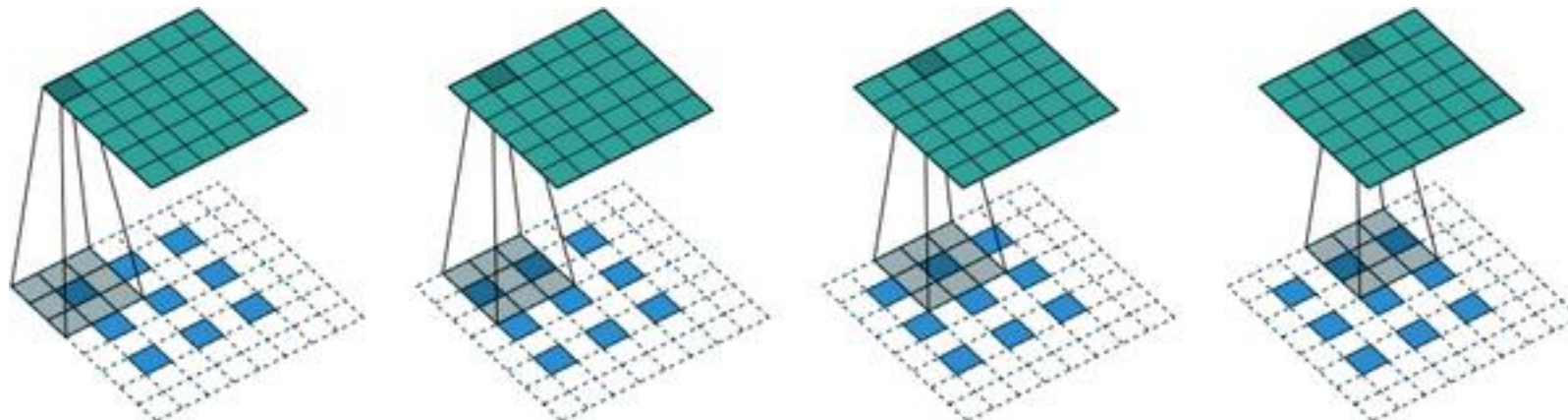
» Train two parts alternatively, only update one network at a time



Convolutional Transpose Layer



- » Sometimes called a "deconvolution" or "up-convolution"
- » A layer that makes an image bigger, by learning how to fill in new pixels, based on patterns it knows
- » It reverses what a regular convolution does:
 - A convolution shrinks the spatial size while extracting features
 - A convolutional transpose expands the spatial size — it upsamples the feature maps to a bigger image





GAN Training Tips and Tricks

- » Notoriously difficult to train .
- » Discriminator overpowers the generator
 - Discriminator becomes too strong, perfectly learns to separate real images from fake images, leading to no training improvements in the generator
 - Make the discriminator “weaker”, add some noises or errors —
- » Generator overpowers the discriminator
 - Known as model collapse, the generator can easily trick the discriminator —
 - Strengthen the discriminator



Label Smoothing (Optional)

- » A useful trick when training GANs
- » Add a small amount of random noise to the training labels —
- » To tame the discriminator, presented with a more challenging task so it does not overpower the generator



Improvements: WGAN (Optional)

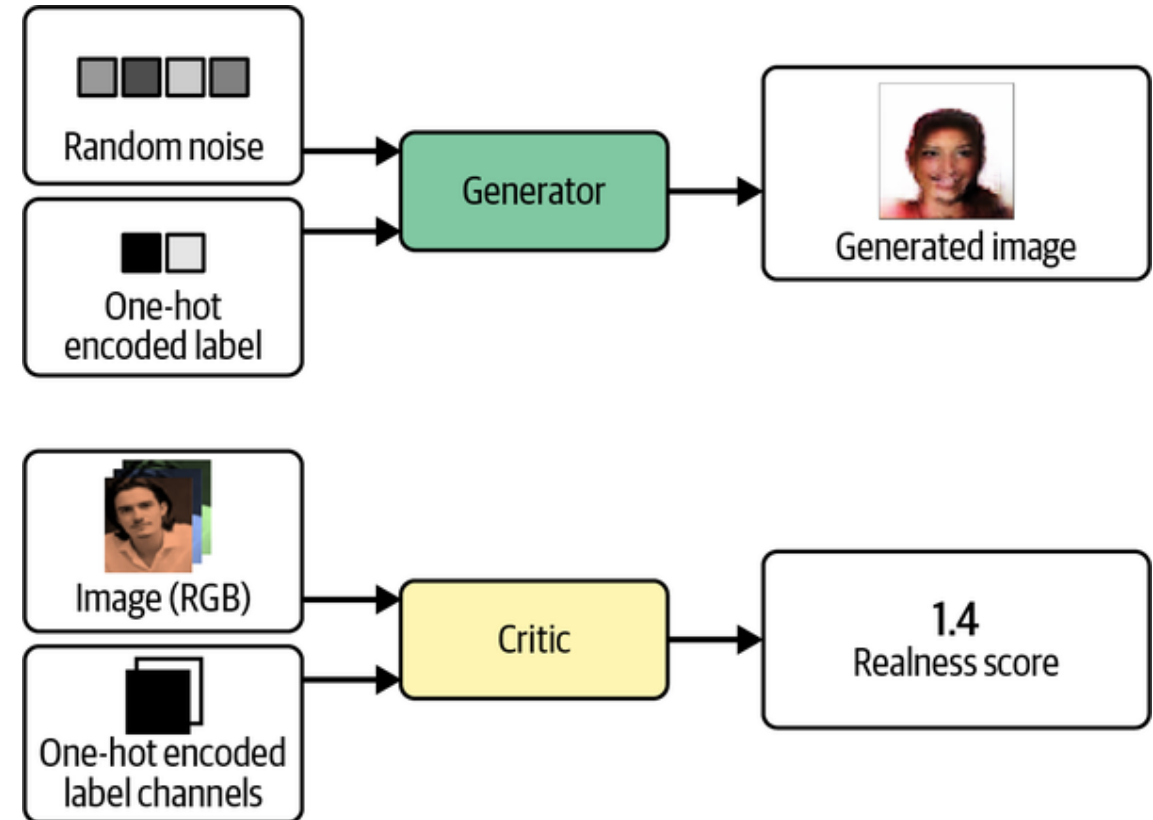
» **Wasserstein GAN with Gradient Penalty (WGAN-GP):** 2017 by Arjovsky et al.

- Stabilize GAN training by using Wasserstein loss function
- Discriminator outputs a score $(-\infty, \infty)$ rather than a probability $[0,1]$, becomes a 'critic'
- Generator tries to produce images that are scored as highly as possible by the critic
- Gradient penalty: penalize the model if the gradient norm deviates from 1

Improvements: Conditional GAN (Optional)



- » Condition the output on attributes, such as “blond hair”
 - Generate an image with blond hair or not
- » Pass in extra information to the generator and critic relating to the label



GAN Usages



- » Virtual try-on
- » Create art pieces
- » Animation



<https://github.com/rkjones4/GANGogh>



GAN Zoo Repository

» <https://github.com/hindupuravinash/the-gan-zoo>

» Over 500 examples of GANs

- ProGAN, StyleGAN, StyleGAN2
 - From NVIDIA
- Self-Attention GAN (SAGAN) and BigGAN
 - GAN with attention
- VQ-GAN, ViT VQ-GAN
 - Blending VAE, transformers, and GANs
 - ViT stands for Vision Transformer

Diffusion vs GAN



- » Diffusion models now outperform previously state-of-the-art GANs
- » Become the go-to choice for generative models particularly for visual domains
- » Introduced in 2020, Denoising Diffusion Probabilistic Model (DDPM)
- » Diffusion: thermodynamic diffusion

Denoising Diffusion Models (DDM)

- » Train a deep learning model to denoise an image over a series of very small steps
 - Forward Diffusion Process: corrupt an image into standard Gaussian noise

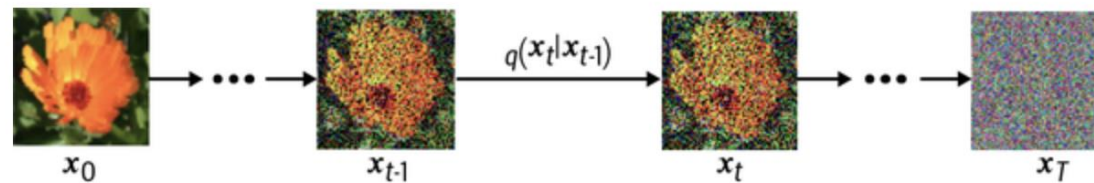


Figure 8-3. The forward diffusion process q

- Reverse Diffusion Process: undo the noising process

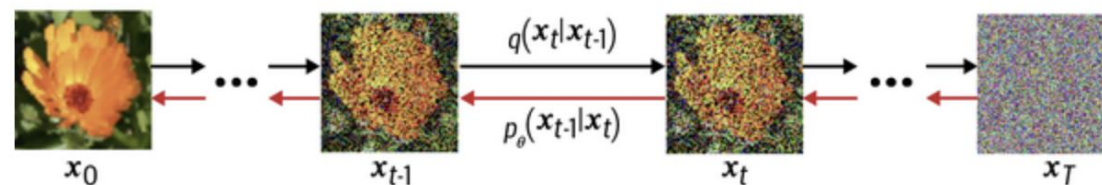
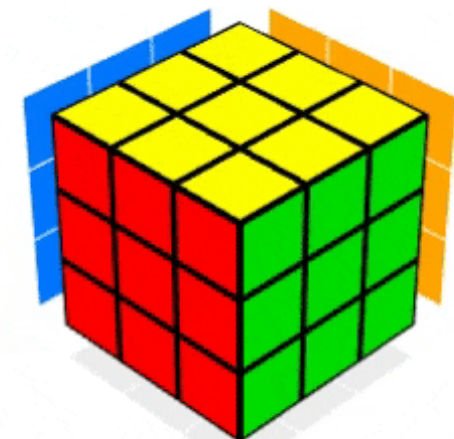


Figure 8-6. The reverse diffusion process p_θ . $(x_{t-1} | x_t)$ tries to undo the noise produced by the forward diffusion process



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URU'R'URU'R'U



<https://medium.com/@krossovochkin/3x3x3-rubiks-cube-world-gif-alert-613d016d2ae7>



Diffusion Schedules

- » Diffusion: modify the pixels mathematically
- » There are different mathematical functions to apply
- » Original paper: linear schedule
- » Cosine and offset cosine schedules are found later to be more effective

Denoising Model: U-Net Architecture

- » Predict the noise added to a given image
- » Downsampling half: compress image spatially, but expand channel-wise
- » Upsampling half: expand representations spatially, but # of channels reduced
- » Skip connections: between equivalent spatially shaped layers in the upsampling and downsampling parts

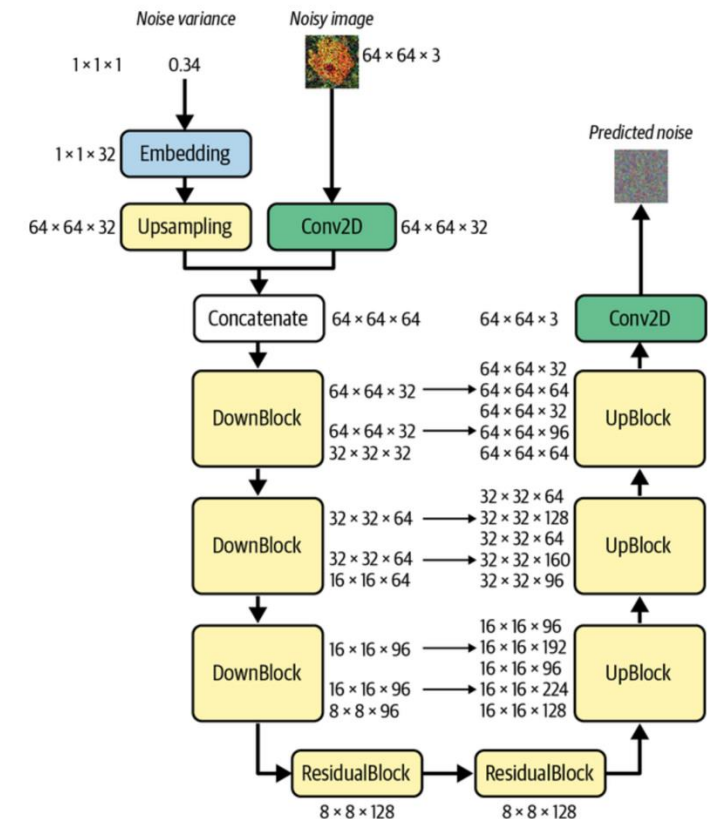


Figure 8-8. U-Net architecture diagram

Residual Block and Sinusoidal Embedding (Optional)



- » A groups of layers that contains a skip connection that adds the input to the output, a highway
- » A solution to degradation/vanishing gradient
 - Vanishing gradient (in deep learning course): as the network gets deeper, the gradient propagated through deeper layers is vanishing thus learning is slow
- » Bypass the weight updates, allows the network to be trained to great depth without sacrificing gradient size
- » Sinusoidal embeddings give the diffusion model a "sense of time," so it knows how much noise to remove at each step!

Multimodality Models



Multimodal Learning

- » Multimodal learning: training generative models to convert between **two or more** different kinds of data
- » Text-to-image: produce images from a given text prompt
- » DALL.E from OpenAI ✓
- » Imagen from Google Brain ✓
- » Stable Diffusion from Stability AI ✓
- » Flamingo from DeepMind ✓

Overview



Table 14-2. Text-to-image models

Model	Date	Developer	# parameters	Open source
DALL·E 2	Apr 2022	OpenAI	3,500,000,000	No
Imagen	May 2022	Google	4,600,000,000	No
Parti	Jun 2022	Google	20,000,000,000	No
Stable Diffusion	Aug 2022	Stability AI, CompVis, and Runway	890,000,000	Yes
MUSE	Jan 2023	Google	3,000,000,000	No



Challenges in Text-to-Image

- » Two tasks: text understanding and image generation
- » As well as how to connect these two
- » Key idea: learn the shared representation
 - Remember the encoder-decoder mechanism
 - Diffusion as the decoder
 - Encoder needs to convert text embedding into visual embedding

Text Prompt → [Transformer Prior] → Latent Code → [Diffusion Decoder] → Final Image

DALL.E Series

- » DALL.E, first version, released in February 2021
- » DALL.E 2 released in April 2022
- » DALL.E 3 initial release in August 2023

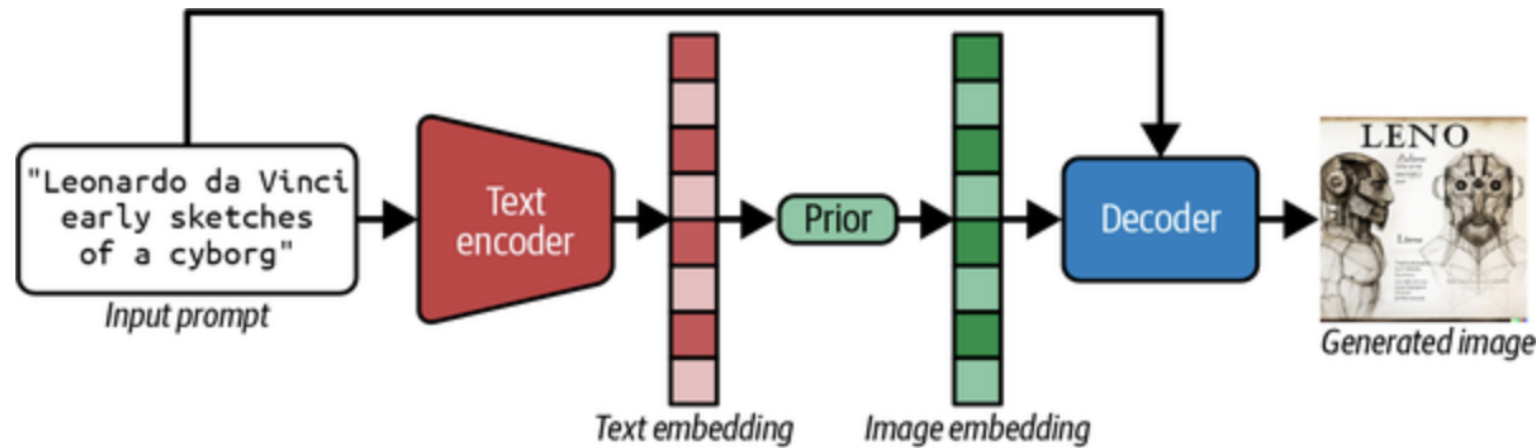


Figure 13-2. The DALL.E 2 architecture

Connecting Text and Image: CLIP



- » Contrastive Language-Image Pre-training (CLIP)
- » A neural network learns visual concepts from natural language supervision
- » <https://openai.com/index/clip/>
- » Published in February 2021
- » Trained on 400 million text-image pairs

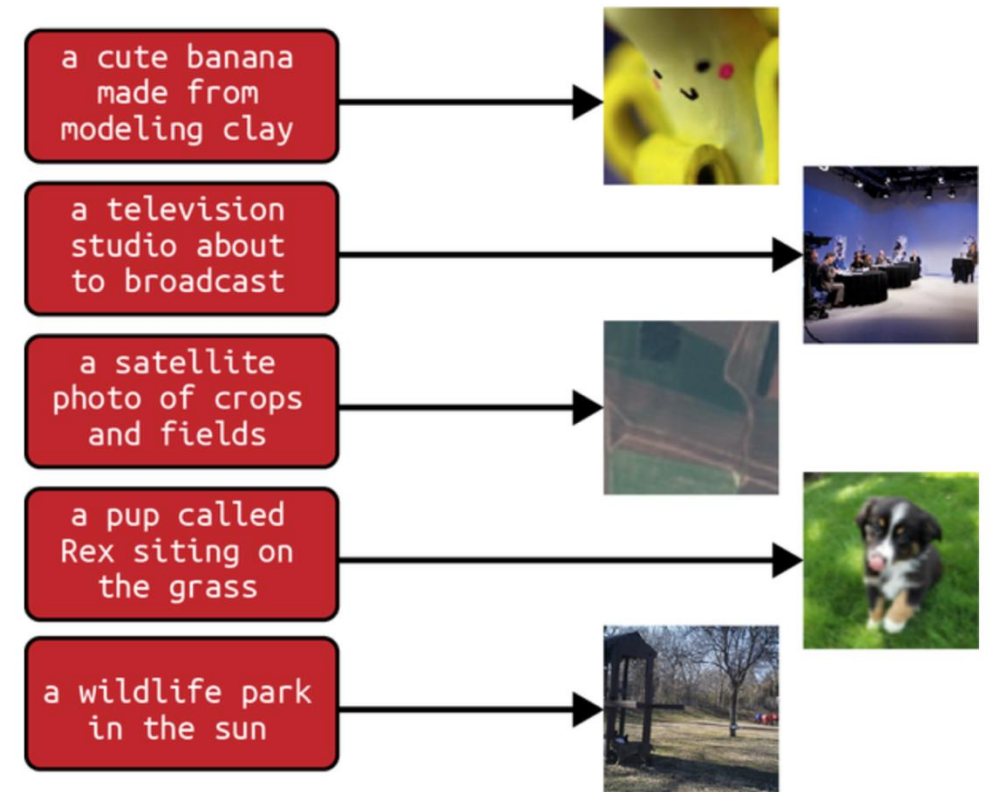


Figure 13-3. Examples of text-image pairs

Prior Model



- » Prior model to convert text embedding into CLIP image embedding
- » DALL.E 2 —
 - OpenAI experimented with both a transformer prior and a diffusion prior
 - And found diffusion approach is better in terms of performance and efficiency
- » DALL.E 3 —
 - OpenAI returned to transformer prior
 - Likely for better integration with ChatGPT



Decoder: GLIDE

- » Guided Language to Image Diffusion for Generation and Editing (GLIDE)
 - Published in December 2021 by OpenAI
 - <https://arxiv.org/abs/2112.10741>
- » A diffusion decoder

Stable Diffusion



- » Specific, optimized diffusion model introduced by Stability AI in 2022
 - Based on the paper “High-Resolution Image Synthesis with Latent Diffusion Models”
 - <https://arxiv.org/abs/2112.10752>
 - Make the diffusion more lightweight, boost speed and performance in training
- » Open-source!
 - <https://huggingface.co/stabilityai/stable-diffusion-3-medium>

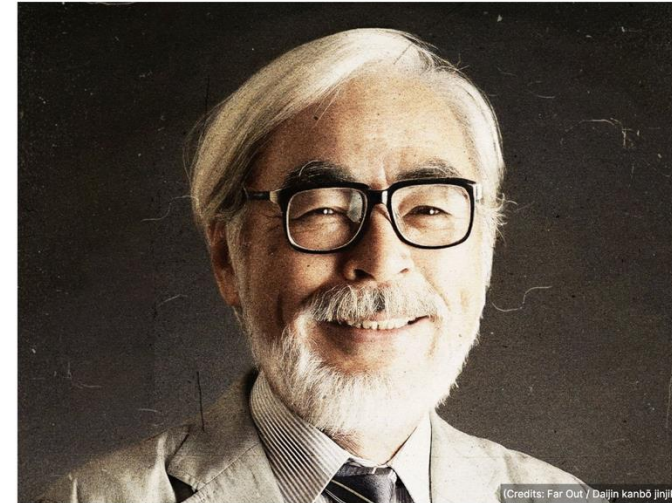
Concerns and Dark Side

Impact on Art



» <https://faroutmagazine.co.uk/hayao-miyazaki-on-ai-utterly-disgusted/>

» <https://www.the-independent.com/arts-entertainment/films/news/hayao-miyazaki-studio-ghibli-ai-trend-b2723358.html>



FILM > CUTTING ROOM FLOOR

Hayao Miyazaki on the use of AI: “I am utterly disgusted”

Tom Leatham

Fri 5 May 2023 17:30, UK

Hayao Miyazaki’s ‘disgusted’ thoughts on AI resurface following Studio Ghibli trend

‘Whoever creates this stuff has no idea what pain is whatsoever’ said Miyazaki

Greg Evans • Friday 28 March 2025 10:05 EDT • 3 Comments



DeepFakes



TECH

Teens are using AI to create fake nudes of their classmates — as a new form of bullying

By [Marissa Matozzo](#)

Published April 21, 2025, 6:01 p.m. ET

18 Comments

Alarming studies reveals rising threat of deepfakes and youth sharing explicit photos for payment

The research also revealed concerning statistics about online solicitations, with over one-third (36%) of young people reporting they had been asked to share sexual imagery of themselves by someone they only knew online while under 18.

BY LOUIS AMESTOY • [NEWS, PUBLIC SAFETY](#) • APRIL 24, 2025

April 25, 2025

Deepfake content presents growing cybersecurity threat

UCalgary's cybersecurity director explains the risk and outlines how you can stay safe online

Potential for deepfake audio, video to warp reality is building to crisis level

Persona warns variety of threats redefining trust

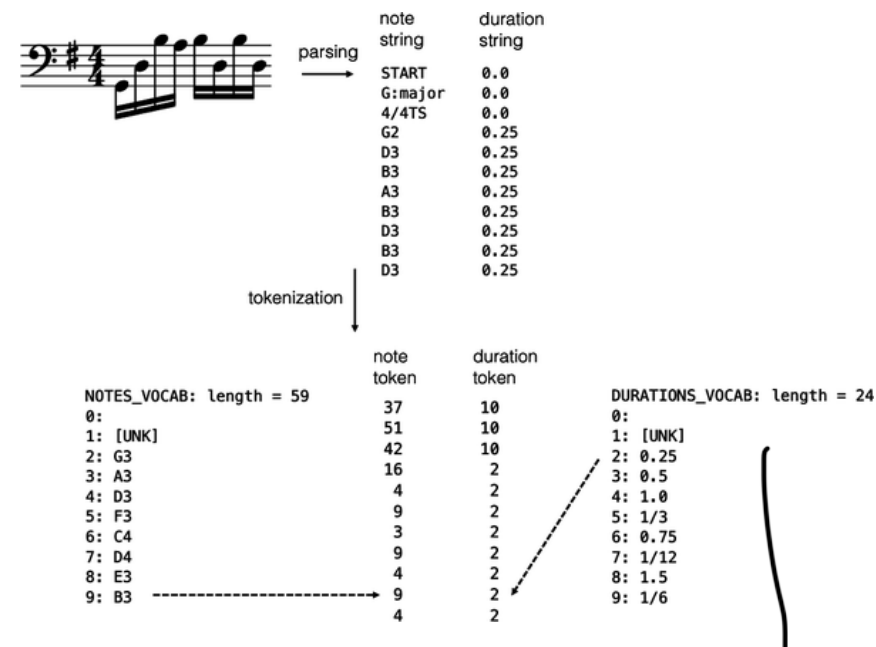
🕒 Apr 21, 2025, 4:47 pm EDT | [Joel R. McConvey](#)



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Music Generation (Optional)

Transformers for Music



Notes string

START G:major 4/4TS rest B3 B3 B3 A3 G3 F#3 G3 D3 E3 F#3 G
3 A3 B3 C4 D4 B3 G3 F#3 G3 E3 D3 C3 B2 C3 D3 E3 F#3 G3 A3 B
3 C4 A3 G3 F#3 G3 E3 F#3 G3 A2 D3 F#3 G3 A3 B3 C4 A3 B3 ...

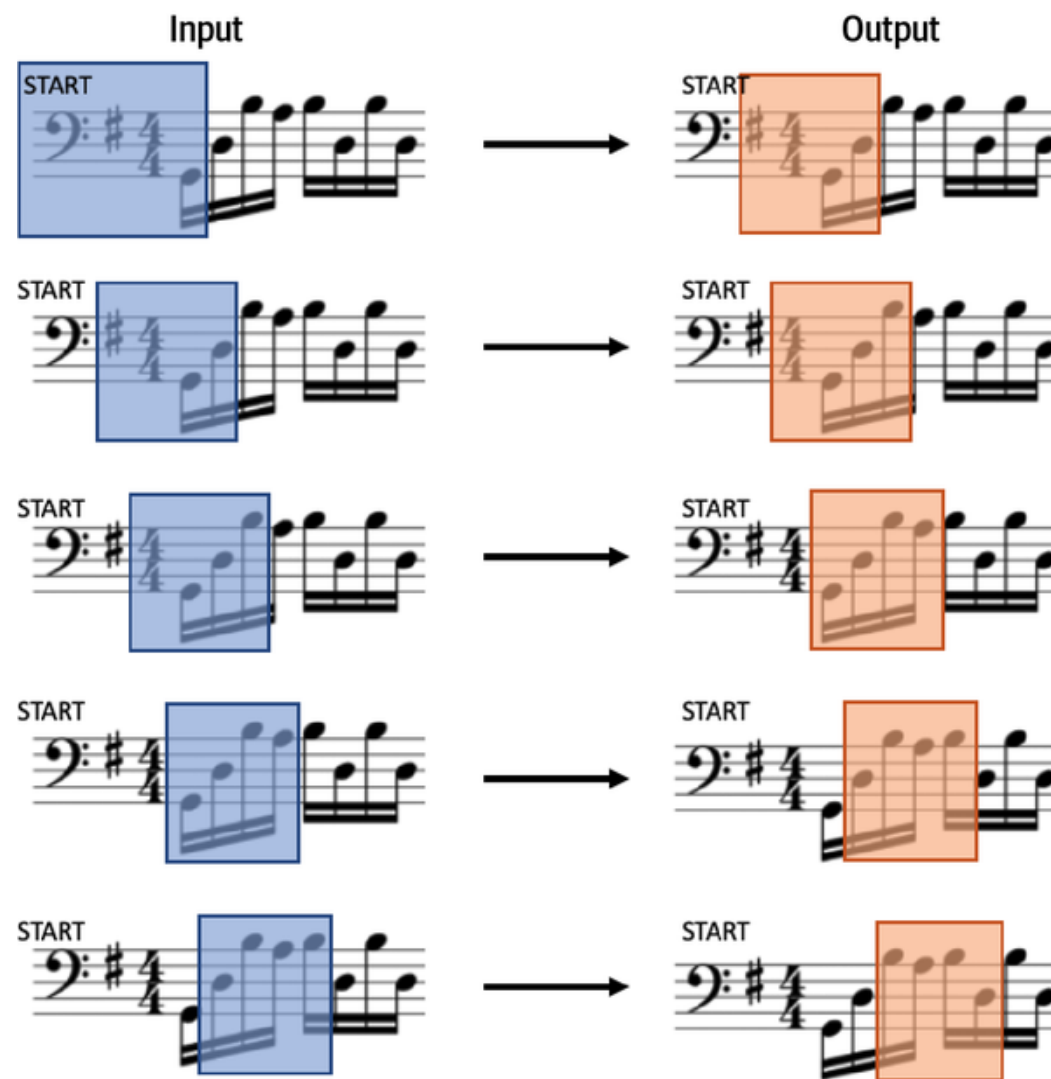
Duration string

0.0 0.0 0.0 3.75 0.25 1.0 0.25 0.25 0.25 0.25 0.25 0.25 0.
25 0.25 0.25 0.25 0.25 0.25 0.25 0.25 0.25 0.25 0.25 0.25
0.25 0.25 0.25 0.25 0.25 0.25 0.25 0.25 0.25 0.25 0.25 0.25
0.25 0.25 0.25 0.25 0.25 0.25 0.25 0.25 0.25 0.25 0.25 0.25
0.25 0.25 0.25 ...

Note: all figures in this section are from
Generative Deep Learning, 2nd edition

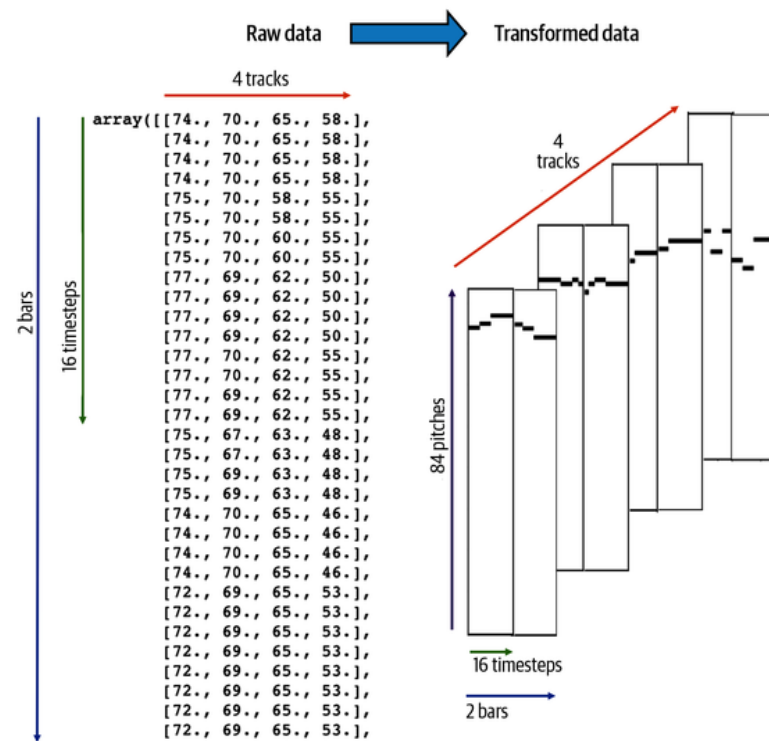
Training

- » Similar to text
- » Output is simply the input window shifted by one note
- » Train to predict the note and duration of the element one timestep into the future
- » <https://magenta.tensorflow.org/music-transformer>

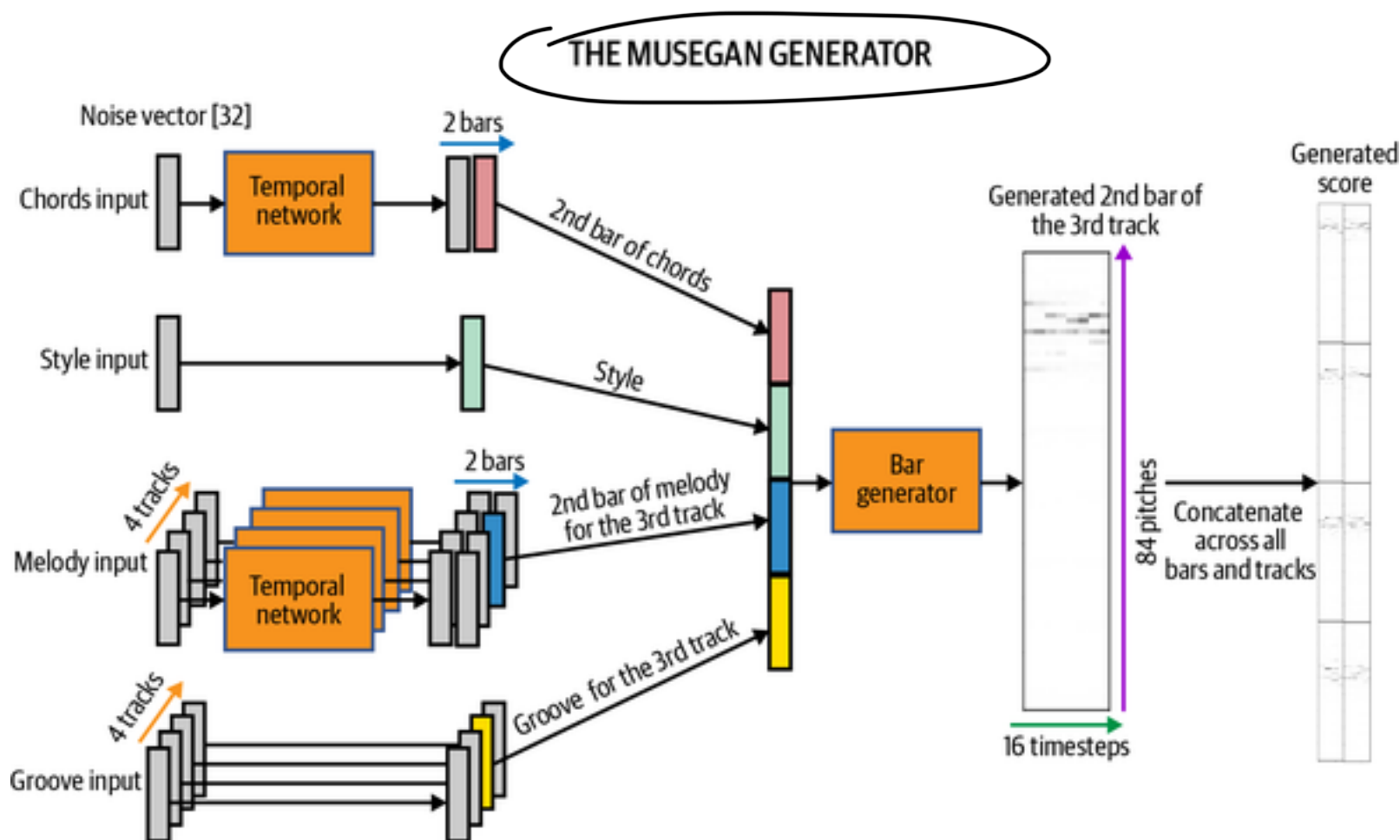


MuseGAN

- » A totally different approach
- » Generate entire track at once
 - Treat piano roll as an image
 - Compared to transformers generating note by note
- » Introduced in 2017, “MuseGAN: Multi-Track Sequential Generative Adversarial Networks for Symbolic Music Generation and Accompaniment”



Architecture





Next Week

»» Project Presentation

- Every team has 12 minutes for presentation
- No reserved Q&A
- Presentation sequence

»» Feel free to invite your collaborator

- Zoom link to join: <https://jhucarey.zoom.us/my/minghong>

»» Final review



Presentation Equipments

- » Recommend to use your own device with HDMI input
- » If your team does not have HDMI-adaptable laptop, wireless presentation can be another option
- » Wireless presentation instructions are posted under Module 6
- » Microphone is available
- » Pointer can be requested from instructor

References



- » Generative Deep Learning, 2nd edition, by David Foster; Publisher: O'Reilly Media (2023)